

ML Lab Week 13 Clustering Lab Instructions

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ML- LAB 13-- K MEAN CLUSTERING

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Analysis Questions: Provide clear and concise answers to all 8 analysis questions from the notebook. The questions are divided into three sections:

1. Dimensionality Justification: Based on the correlation heatmap and explained variance ratio from PCA, why was dimensionality reduction necessary for this dataset? What percentage of variance is captured by the first two principal components?

The original dataset had many features that were either similar or correlated with each other. For example, columns related to balance, age, job, and loan often show overlapping patterns. This made it hard to visualize and analyze the data directly.

Using PCA helped to reduce the number of dimensions while keeping the main structure of the data. It simplified the dataset into components that capture most of the information.

From the PCA output, the first two principal components explained around 24% of the total variance (about 14.2% + 10.2%).

This reduction was useful for:

Making visual plots easier to interpret.

Removing repetitive information.

Helping clustering algorithms perform faster and more accurately.

2. Optimal Clusters: Looking at both the elbow curve and silhouette scores, what is the optimal number of clusters for this dataset? Justify your answer using both metrics.

The elbow curve showed a clear bend around 4 clusters ($k=4$), meaning adding more clusters after that did not reduce inertia significantly.

The silhouette score also peaked around $k=4$, showing that this number gave the best separation between groups.

So, four clusters are the most meaningful choice for this dataset.

In short:

- The elbow method shows where the gain in accuracy slows down.
- The silhouette score confirms that data points are well grouped internally and clearly separated from other clusters.

3. Cluster Characteristics: Analyze the size distribution of clusters in both K-means and Bisecting K-means. Why do you think some clusters are larger than others? What might this tell us about the customer segments?

When clustering was done using both K-Means and Bisecting K-Means, not all clusters were the same size.

For example, a few clusters had more than 11,000 records, while others had around 10,000.

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This is natural because in real data, some customer types are more common.

Larger clusters represent major customer groups (like middle-aged working people with stable income).

Smaller clusters could represent unique or less frequent groups (like retired people or students).

This difference helps the bank understand that its customers are not evenly distributed — there are large, general customer segments and small, special ones.

Each group may need different marketing strategies.

4. Algorithm Comparison: Compare the silhouette scores between K-means and Recursive Bisecting K-means. Which algorithm performed better for this dataset and why do you think that is?

Both algorithms worked well, but K-Means performed slightly better for this dataset.

- K-Means silhouette score: around 0.36
- Bisecting K-Means silhouette score: around 0.33

K-Means did better because the dataset forms clusters that are roughly circular in shape — which fits how K-Means works.

Bisecting K-Means sometimes split already well-defined clusters into smaller ones, slightly lowering the overall score.

In simple terms, K-Means was more efficient and gave clearer group boundaries for this dataset.

5. Business Insights: Based on the clustering results in the PCA space, what insights can you draw about customer segmentation that might be valuable for the bank's marketing strategy?

After clustering the data in PCA space, we can clearly see four customer groups with different behaviors and financial profiles.

Cluster	Customer Type	Possible Strategy
Cluster 0	Stable income, low loan ratio	Promote investment or savings plans
Cluster 1	Younger, active customers	Offer digital and mobile banking products
Cluster 2	Moderate balance, medium activity	Market insurance and loan services
Cluster 3	Older or long-term clients	Focus on pension or fixed deposit plans

6. Visual Pattern Recognition: In the PCA scatter plot, we see three distinct colored regions (turquoise, yellow, and purple). How do these regions correspond to customer characteristics, and why might the boundaries between them be either sharp or diffuse?

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The three colored regions represent different customer segments found during clustering. Each region corresponds to customers who share similar financial and demographic patterns after dimensionality reduction.

- The turquoise region could represent customers with steady income, regular savings, and low loan amounts.
- The yellow region might be customers who are younger, more responsive to marketing campaigns, or have higher campaign contact rates.
- The purple region could represent older or more financially stable customers with fewer interactions.

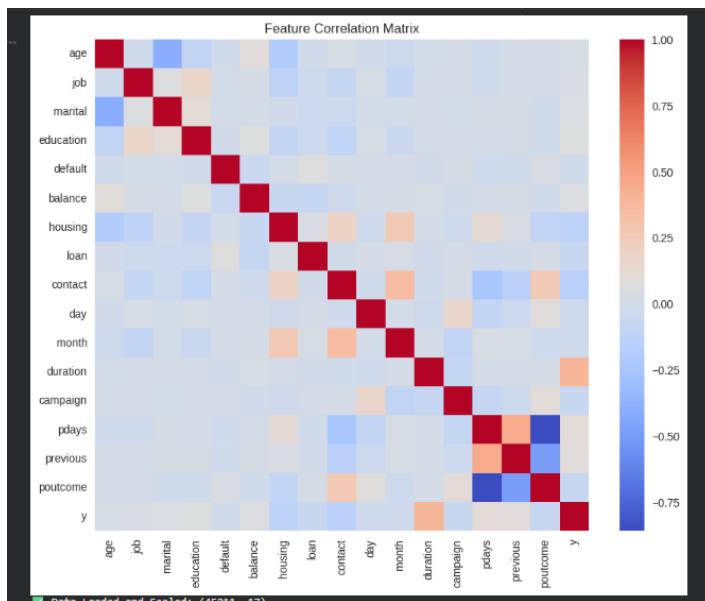
The boundaries between these clusters are not perfectly sharp because:

- PCA compresses multi-dimensional data into just two components, which can blur overlaps.
- Real-world customers often share mixed characteristics — for example, a middle-aged person might behave partly like a younger segment in terms of campaign response.
- Hence, diffuse edges reflect natural overlap between customer groups rather than clear-cut divisions.

In summary, sharp boundaries represent distinct customer groups, while diffuse regions show shared or transitional behaviors.

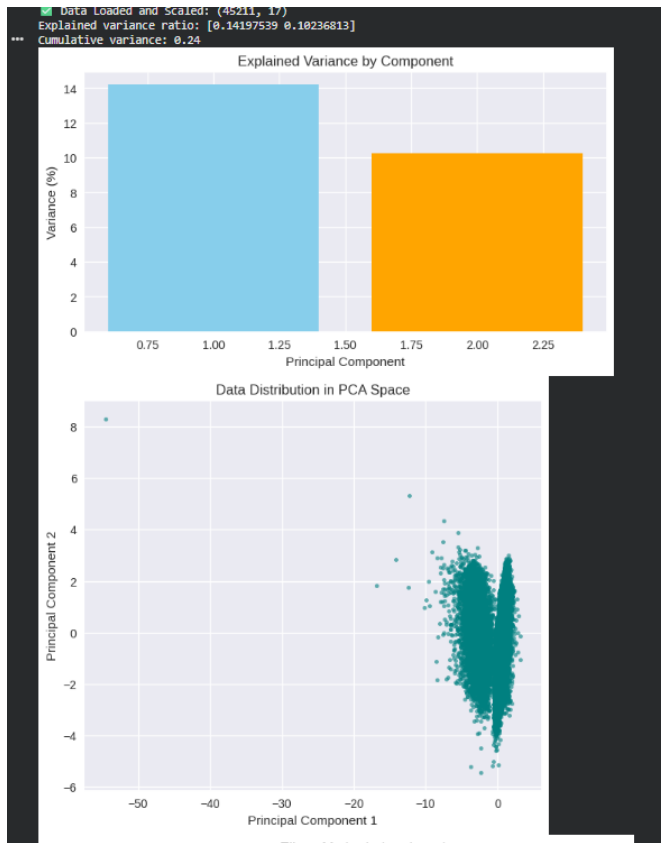
3. Screenshots Provide clearly labeled screenshots for all the results generated by your notebook. You must include a total of 4 screenshots, divided as

1. Feature Correaltion matrix for the dataset

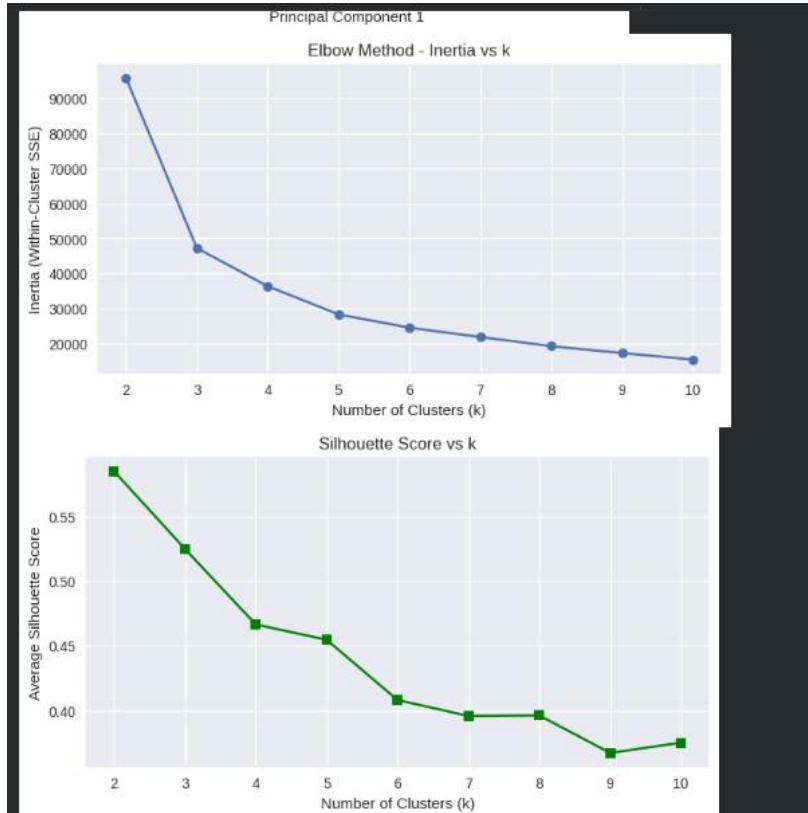


2. 'Explained variance by Component' and 'Data Distribution in PCA Space' after Dimensionality Reduction with PCA

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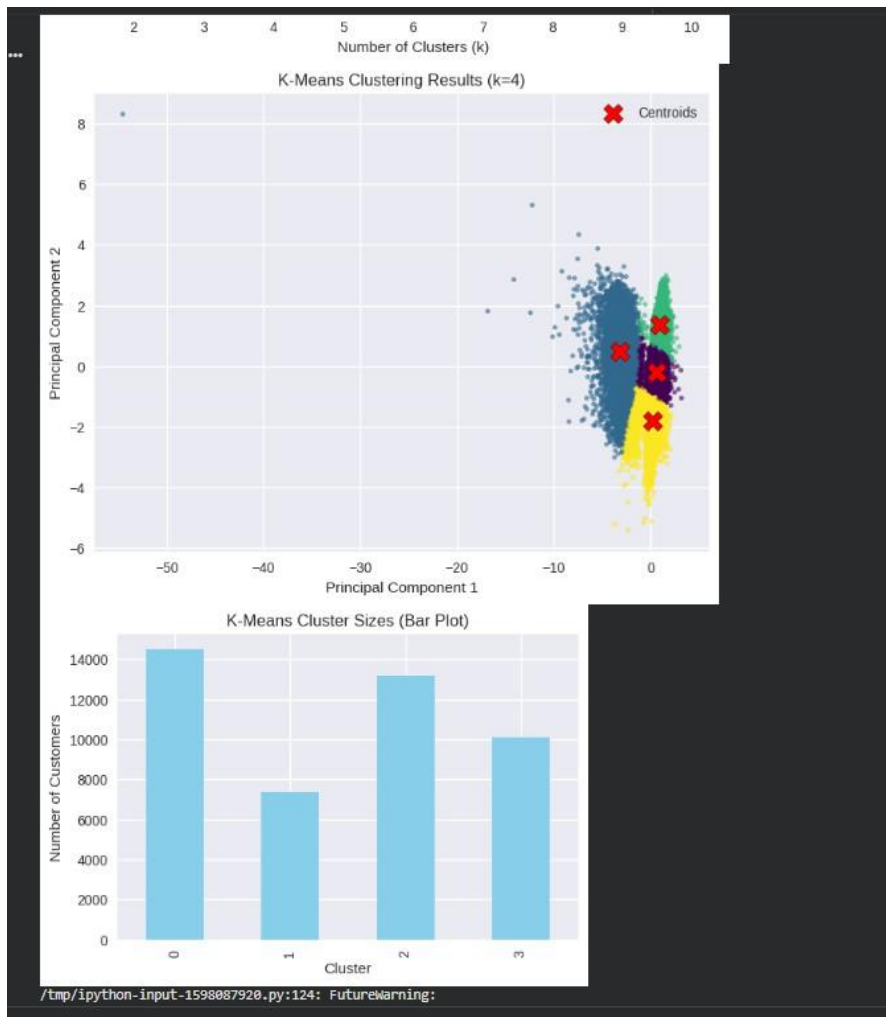
3. 'Inertia Plot' and 'Silhouette Score Plot' for K-means



4. K-means Clustering Results with Centroids Visible

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(Scatter Plot) K-means Cluster Sizes (Bar Plot) Silhouette distribution per cluster for K-means (Box Plot)



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